

Choosing the lamp for 410–680 nm

A spectral evaluation of Avonec 3 W LED combinations for pumpkin-seed-oil measurement, including the deep-red star.

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1 · The question, and the short answer

Edwin asked for the Avonec LED combination that gives the best spectrum for the pumpkin-oil measurement **across 410–680 nm**, taking into account a **3 W star-mounted 660 nm emitter**.

This document answers it by scoring **43 650 buildable seven-emitter allocations** against the pigment's own absorption bands, computed from the **measured** spectra Avonec publishes for its own parts.

★ **680 nm is not an arbitrary top.** It is exactly where KB_spectroscopy_physics.md §7.2 puts the last measurable response — the Eu³⁺ 650.7 nm line resolves and response continues "to ~680 nm". So the range asked for and the range the instrument can deliver are, for once, the same interval.

Five findings. The first answers the star question; the fourth is a correction to this document itself.

	finding
1	★★ You do not need a second vendor. Avonec sells the 660 nm on a star board itself — <i>3W High Power LED auf Starplatine 660nm tiefrot/hyper red, €2.75</i> , 2–4 working days, and its measured spectrum is already on file. A second vendor is needed only for 670/680 nm, which Avonec does not sell.
2	★ The backbone changes. Over 410–680 nm the best white is 4000–4500 K , not the 6500–7000 K the 430–670 study chose (50.12 against 55.71 alone). It is the only white with both a usable blue pump <i>and</i> a real red tail.
3	★ The 410 nm end is the bargain. One 410–420 nm emitter lifts 410 nm from 0.036 to 0.44 (12×) and the sub-Soret level 2.7× . It is the largest single gain in the study.
4	⊖ An earlier version of this document claimed the camera ends the range past 640 nm. That claim is WITHDRAWN (§6) — the ~40× fall it rested on was measured through a CFL, whose red output is Eu ³⁺ <i>line</i> emission, so the ratio is the lamp's own structure. ★ Working assumption now: the optics deliver to ~700 nm and the camera captures it , so nothing inside 410–680 nm is known to be detector-limited.
5	★ The forecast says every region of 410–680 nm is usable (§8). Modelled green and brown oils under the recommended lamp put nothing below the 1 % floor — and the quiet window 660–680 nm arrives essentially undiminished, because the oil is transparent there.

⇒ **The recommendation:** build **R2** (§7.1) — all Avonec, all on Starplatine — and add a 670 nm star if the quiet window is wanted now (§7.3). ★ **Neither half has to be taken on trust first:** §7.4 shows how to price both — the red half free today on the existing lamp, the blue half with one emitter.

★ **And the concrete benefit, replacing §6's withdrawn one (§6.3): the 619/624 nm red feature** is real, moves between runs, and sits on the far anchor 620–630 where errors carry 17× leverage. It is present on *both* bench lamps — the Sansi's 619 nm edge **amplifies it 4–6×**, **it does not cause it**. R2 reads 0.1–0.2 %/nm across that band. ⇒ **the recommended lamp removes the multiplier.**

△ **What this study is.** It ranks *candidate spectra*. It knows nothing about drive current, binning, thermal droop, diffuser transmission, or how emitters combine behind one diffuser — and real per-part output varies far more than normalised curves suggest. Treat every ranking here as a **shortlist for the bench**, not a result. §8's oil forecast is a *forward model* on top of that and carries its own, larger, uncertainty — §8.2 states the check it fails.

1.1 ⚠ Two corrections to the first draft of this study

Both were found by checking the shop rather than the archive, and both matter.

★★ the 660 nm star is an Avonec part	The first draft treated it as a second-vendor item with no measured spectrum, and modelled it. It is Avonec's own, so the measured <code>660nm.jpg</code> curve applies. §5.2's conclusion survives — but for a better reason than it had.
★ Avonec does sell a part inside the cyan gap	The first draft said " <i>nothing between 460 and 515 nm</i> ". Wrong: there is a 480–485 nm türkis (€2.75, on Starplatine) and a 490–495 nm cyan (sold out). Its Spektralmessung was harvested on 2026-08-07 and is now scored — see §4.2 finding 2.

2 · Why 410 and why 680 — the two ends are not symmetric

The range was not chosen for roundness. Each end answers a different question about the pigment, and the two questions have very different price tags.

2.1 The blue end: 410–430 nm makes the Soret a *band* instead of a flank

The oil's dominant absorption is the **Soret band of protochlorophyll at ~432 nm** (`KB_spectroscopy_physics.md` §4.1). The shipped measurement window is **448–460 nm** — 16 to 28 nm above the peak, i.e. **pure flank**. It sits there because that is where the instrument had light, not because that is where the chemistry is.

★ **And the chemistry that matters is a *shift*, not only a depth.** Roasting and ageing strip the central Mg²⁺ from protochlorophyll, giving **protopheophytin**. Demetallation lowers the macrocycle's symmetry, and the textbook consequence is that **the Soret weakens and blue-shifts**. That is the green → brown axis at the molecular level.

⊖ **A blue shift cannot be measured from one flank.** Seen only from 448–460 nm, a band that moves left and a band that gets weaker look identical. To separate them you need light on **both sides** of 432 nm — which is exactly what 410–430 nm buys.

⇒ The objective therefore scores not just the photon count at 432 nm but the **median level across 415–450 nm**, the *bracket*.

⚠ **One caveat on that bracket, found while checking the results.** 415–450 nm includes 450 nm, where every phosphor white has its blue *pump peak* — so a lamp can score a good bracket on light that sits entirely **above** the Soret and still be blind below it. The incumbent does exactly that (bracket 0.712, propped by a pump peak of 1.000 at 450 nm, while reading **0.036 at 410 nm**). ⇒ **The honest discriminator is the level strictly below the peak**, reported throughout as *sub-Soret* — the median across **415–428 nm**. It is a reported diagnostic, not part of the objective, which stays frozen at the bracket the ranking was computed on.

2.2 The red end: 660–680 nm is the first pigment-free window

Since the 2026-07-31 correction, the oil's red band is known to be the **Qy of protochlorophyll at ~623–626 nm**, not chlorophyll a's 665 nm. That correction has a direct consequence for the range:

★ **Everything above ~660 nm is genuinely quiet.** There is no pigment feature there. That makes 660–680 nm the **baseline anchor the metric has never had** — the far anchor in use today (620–630 nm) *contains the Qy band itself*, which is the whole reason `DOC_pedestal_correction.md` exists and why the chord over-subtracts. A truly pigment-free anchor would make that correction unnecessary rather than unproven.

⇒ So the red end is worth more than the blue end *scientifically*. §5 and §6 show it is also far more expensive.

3 · Method

3.1 The source data — Avonec's own measured spectra

Avonec publishes a *Spektralmessung* for each 3 W part, but as a **JPG plot, not as numbers**. The set is harvested to `spectracs-references/leds/avonec/`. This study digitises the plotted curve: the x axis is calibrated on the plot's gridlines (with the image borders discarded, since the white margin also reads as grey), and the y axis needs no calibration because every Avonec plot is normalised to 1.00 at its own peak.

★ **Thirteen curves are now calibrated, against seven in the 430–670 study.** That study excluded four colour parts rather than guess their axes; those axes have now been read off the plots, the two remaining whites with them, and the **480–485 nm** part was harvested for this study — it had been missed entirely.

Every digitised peak is checked against the part number (--verify):

plot	digitised peak	part says	
410nm-420nm	421.0 nm	421	✓
430nm-435nm	430.0 nm	432	✓
440nm-450nm	441.5 nm	441	✓
455nm-460nm	457.5 nm	458	✓
480nm-485nm	480.5 nm	481	✓ harvested 2026-08-07
515nm-525nm	514.5 nm	520	✓
590nm-600nm	594.0 nm	594	✓
600nm-610nm	615.0 nm	615	✓ ⚠ named 600–610, measures 615
630nm-640nm	635.5 nm	635	✓
660nm	660.5 nm	661	✓
the five whites	—	bimodal, no single peak	—

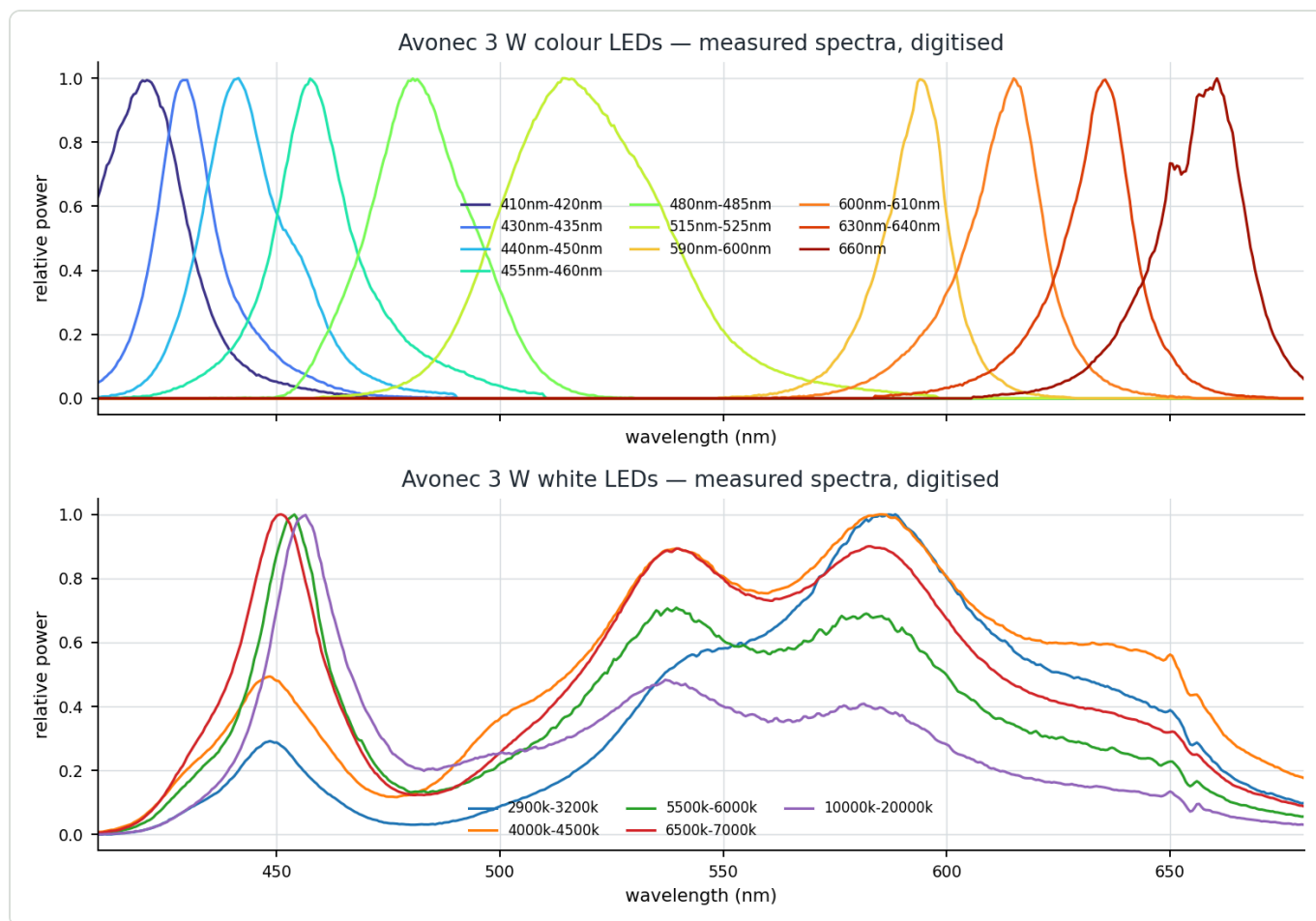


Figure 1 — Every Avonec 3 W spectrum used in this study, digitised from the shop's own Spektralmessung plots. Top: the ten colour parts, including the 480–485 nm türkıs harvested for this study. Bottom: the five whites — each a narrow blue pump plus a broad phosphor hump, with the cyan dip near 480 nm common to all of them.

3.2 What is being scored

⊖ Not flatness. The measurement windows sit on flanks (§2.1), and a source's **peak must sit where you measure, never its flank** — at a maximum, sensitivity to wavelength drift is second-order; on a flank it is first-order. Measured three ways on the rig: an emitter *edge* inside an anchor costs 25 %/nm, an emitter *centred* on it costs 0–5 %/nm.

The objective, in order:

	critterion	why
1	photons at 432 / 574 / 625 nm	the pigment's own band centres
2	median level across 415–450 nm (Soret bracket) and 660–680 nm (quiet window)	the two things the widened range exists to buy
3	the log-slope $\text{abs}(d\ln I/d\lambda)$ inside every measurement band, Q-band and far anchor weighted $\times 4$	§16.24.2's 17× error asymmetry — a steep emitter edge inside a band amplifies every R → S mismatch
4	no hole across 410–680 nm	the retired DIY array's 3× cliff at 500 nm is the failure mode to avoid

3.3 The candidate set — integer emitters, not continuous weights

Candidates are **seven emitters allocated as whole parts**, matching the construction sketch: a white phosphor lamp in the centre ringed by ~4 small heat-sinked emitters behind one common diffuser. At least one white; at most three distinct add-on part numbers, so the board stays a board.

⚠ **One assumption is load-bearing and it is not exactly true.** One part is treated as weight 1.0 of its own normalised curve — i.e. every 3 W emitter is assumed to deliver the same radiant flux. It does not: green and cyan emitters sit in the semiconductor **"green gap"** and convert markedly worse than either the blue or the deep-red parts either side of them. ⚠ Avonec's datasheet PDFs are image-only, so per-part figures could not be read here. Counts are a **first-order allocation**, and any green or cyan slot should be read as *"at least this many"*.

3.4 The three families

family	contents	count
A · Avonec only	the catalogue, orderable from one shop today — ★ this now includes the 660 nm star	15 680
B · + modelled 660 star	kept only as a sensitivity check on the measured 660 curve (\$5.3)	20 960
C · + longer deep-reds	adds modelled 670 / 680 / 690 nm parts Avonec does not sell	43 650

⚠ **The 670/680/690 parts are MODELLED** as skew-normal emitters (red-skewed, as real AlGaInP deep-reds are). They have no published measurement on file.

4 · Results

4.1 The ranking, by family

Columns are the emitted level at each wavelength, relative to the combination's own peak. *worst* is the steepest median log-slope over any measurement band, in %/nm. Lower score is better.

Notation. 3×4000K · 2×410 · 1×430 · 1×455 means three 4000–4500 K whites, two 410–420 nm, one 430–435 nm and one 455–460 nm emitter. star670 and up are the parts Avonec does not sell.

A · Avonec only — everything here is orderable today

allocation, 7 emitters	I@432	I@625	I@660	I@680	worst	hole	score
3×4000K · 2×410 · 1×430 · 1×455	0.831	0.598	0.353	0.176	★ 3.7	3.5	18.99
3×4000K · 2×410 · 1×440 · 1×660	0.670	0.612	0.682	0.196	6.8	5.2	19.03
★ 3×4000K · 2×410 · 1×430 · 1×660	0.810	★ 0.612	★ 0.682	★ 0.196	6.8	5.4	19.13
3×4000K · 2×410 · 1×440 · 1×455	0.691	0.598	0.353	0.176	★ 3.0	3.6	19.13
3×4000K · 2×410 · 1×430 · 1×440	★ 0.923	0.570	0.336	0.168	5.3	4.7	19.34
3×4000K · 2×410 · 1×430 · 1×480	0.810	0.598	0.353	0.176	4.2	★ 3.4	19.46

△ **The top six span 18.99 to 19.46 — that is a tie, not a ranking.** What separates them is *which* weakness you accept, not overall quality. The two without a deep red (rows 1 and 4) score marginally better only because a 660 nm part raises the *worst* band slope to 6.8 %/nm; that steepness sits in the **quiet window**, where nothing is measured today, not in a live band. ⇒ **Take the deep red.**

C · + longer deep-reds — the only family that fills the quiet window

allocation, 7 emitters	I@432	I@625	I@660	I@680	worst	score
★ 3×4000K · 2×410 · 1×440 · 1×star670	0.670	0.598	0.541	★ 0.407	5.1	15.97
3×4000K · 2×410 · 1×430 · 1×star670	0.810	0.598	0.541	★ 0.407	5.1	16.10
3×4000K · 1×410 · 1×430 · 2×star670	0.670	0.598	0.729	★ 0.638	5.0	16.28

★ **670 nm, not 680 nm, is the right part for this range** — with the top at 680 nm a 670 emitter sits *inside* the window with its flank covering 660–680, where a 680 emitter wastes half its output past the edge. (In the earlier 689 nm version of this study, 680 won. The 9 nm change moved the answer.)

For reference — the backbones alone, and the incumbent

allocation	I@432	I@574	I@625	I@660	I@680	score
7 × 4000–4500 K	0.231	0.893	0.598	0.353	0.176	★ 50.12
7 × 6500–7000 K	0.288	0.831	0.412	0.197	0.088	55.71
7 × 5500–6000 K	0.190	0.639	0.302	0.128	0.057	80.35
7 × 2900–3200 K	0.094	0.845	0.507	0.226	0.099	143.22

allocation	I@432	I@574	I@625	I@660	I@680	score
7 × 10000–20000 K	☹ 0.099	☹ 0.384	☹ 0.162	☹ 0.070	☹ 0.031	☹ 165.32
incumbent — 3×6500K · 2×430 · 1×515 · 1×660	0.815	0.779	0.393	0.485	☹ 0.099	26.44

★ **The incumbent ranks 1 897th of 43 650 on the widened objective** — not because it was wrong, it won the range it was scored on, but because its two weak points are exactly the two ends this study adds: **0.099 at 680 nm** and a **sub-Soret level of 0.322 against 0.87–0.89**.

4.2 Where the light actually lands

The most useful single table in this study: what each candidate puts on the grid, relative to its own peak. **Every value below 0.10 is in the starved regime that produced the dead 440–447 bins.**

	410	432	450	480	500	520	540	574	625	660	680
R0 incumbent	☹ 0.036	0.815	1.000	△ 0.135	0.416	0.789	0.970	0.779	0.393	0.485	☹ 0.099
R1 Avonec, no deep red	0.443	0.831	0.748	△ 0.182	0.375	0.572	0.892	0.893	0.598	0.353	△ 0.176
★ R2 Avonec 660 star	0.442	0.810	0.561	△ 0.126	0.362	0.572	0.892	0.893	0.612	★ 0.682	△ 0.196
R3 R2 + the 480 filler	△ 0.233	0.670	0.545	★ 0.456	0.475	0.575	0.892	0.893	0.612	★ 0.682	△ 0.196
R4 reach 680 (670 star)	0.427	0.670	0.708	△ 0.133	0.362	0.572	0.892	0.893	0.598	0.541	★ 0.407

	score	rank	sub-Soret 415–428	quiet 660–680	deepest dip	at
R0 incumbent	26.44	1897	☹ 0.322	☹ 0.221	0.066	410 nm
R1 Avonec, no deep red	18.99	106	★ 0.877	0.237	0.282	680 nm
★ R2 Avonec 660 star	19.13	119	★ 0.874	0.352	0.184	476 nm
R3 R2 + the 480 filler	19.09	114	△ 0.544	0.352	★ 0.315	680 nm
R4 reach 680 (670 star)	15.97	1	0.737	★ 0.545	0.212	477 nm

Three things fall out.

★ **1 · The 410 nm gain is the biggest single number in this study.** 0.036 → 0.44, a **12× lift**, and sub-Soret **0.322 → 0.87**, a 2.7× lift. It is attributable to the violet part, not the backbone swap: a bare 4000–4500 K board reads **0.007** at 410 nm; adding one 410–420 nm emitter takes it to **0.112** — 16× from a single part, and the score from 50.12 to 26.46. A 430–435 nm part in the same slot gives a better I@432 (0.380 vs 0.301) but leaves 410 nm at **0.016**: it lights the peak without bracketing it.

△ **2 · The 480 nm hole CAN be filled — but not for free, and that corrects the first draft.** Avonec does sell a **480–485 nm türkis** part (§1.1). Adding it lifts 480 nm from **0.126 to 0.456**, a 3.6× fill of the deepest hole in the spectrum, and it moves the global minimum off 476 nm entirely. ☹ **But on a seven-slot board it**

costs a violet, and the sub-Soret level collapses from **0.874 to 0.544**. The total score barely moves (19.13 → 19.09), which is the honest reading: **it is a trade, not an upgrade**.

⇒ **Recommendation: skip it at seven slots, take it at eight**. The cyan region is not in any measurement band today; the Soret bracket is the whole point of the blue end. △ But it is inside the carotenoid/lutein absorption (440–480 nm), so if a carotenoid feature is ever wanted, this part — not a phosphor — is what makes it measurable.

★ **3 · The green LED disappears from every top allocation**. With a 4000–4500 K backbone the phosphor already delivers 0.89 at 540 nm; a 515–525 nm part adds nothing the score can see. That reverses the 430–670 study's recommendation, and the backbone swap is the reason, not a change of mind about green.

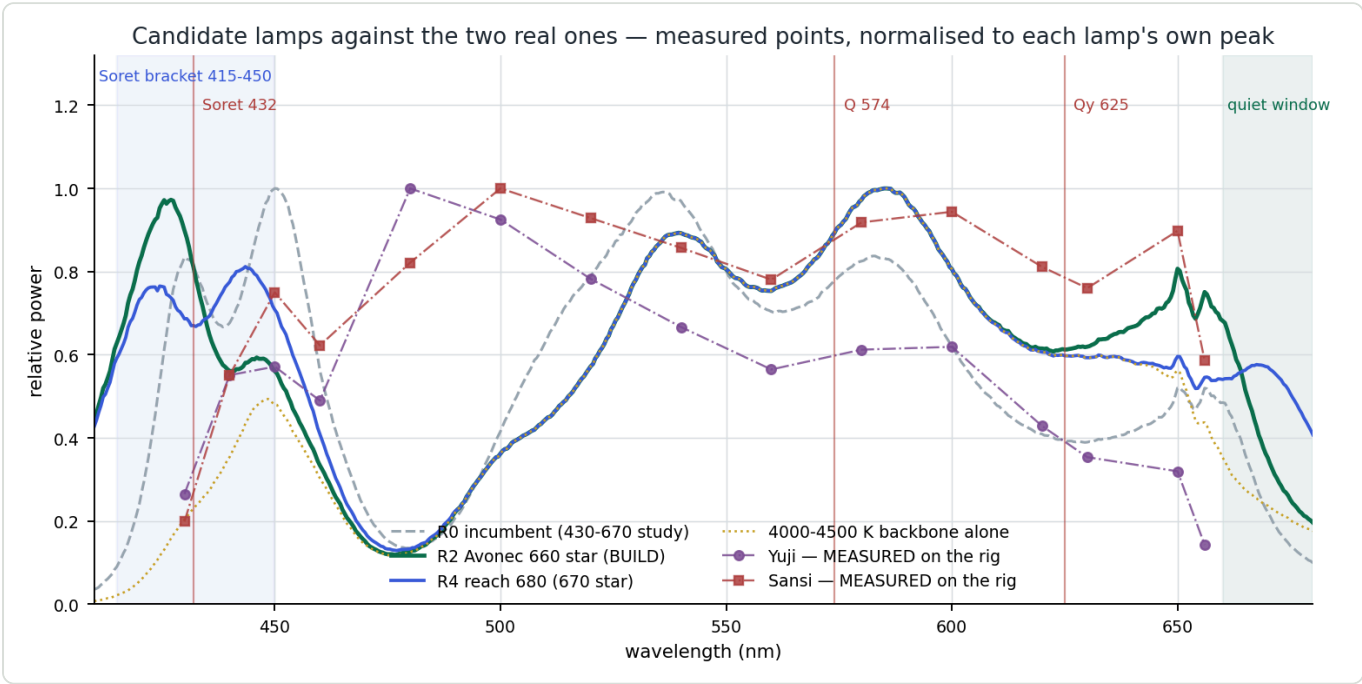


Figure 2 — What the widened range buys, against the incumbent lamp. Grey dashed: the incumbent, blind below 425 nm and above 660 nm. **Solid green: R2, the recommended build** — a broad plateau across the whole Soret bracket and real output at 660 nm. Blue: R4, the 670 nm variant, the only one that fills the quiet window. Dotted gold: the 4000–4500 K backbone alone, showing how much of the middle comes free from the phosphor.

4.3 The ranking is not an artefact of the weighting

The objective is a judgement call, so it was stated five ways and the winner re-derived under each.

weighting	winner	rank of "as written"
as written	3×4000K · 2×410 · 1×440 · 1×star670	1 / 43 650
photons first (band centres ×2)	3×4000K · 2×410 · 1×430 · 1×star670	4 / 43 650
smoothness first (steepness ×2.7)	same as "as written"	1 / 43 650
quiet window ×2.7	2×4000K · 2×410 · 1×440 · 2×star670	10 / 43 650
Soret peak only (bracket removed — the control)	3×4000K · 1×410 · 1×430 · 2×star670	4 / 43 650

★ **First under two weightings, 4th, 4th and 10th under the others**. And note what the control shows: even when the bracket term is deleted — i.e. when the study is reduced back to the 430–670 question — the **410–420 nm part still survives in the winner**. Its value is not an artefact of the criterion invented to justify it.

△ **Two constants are robust across all five: the 4000–4500 K backbone and the 410–420 nm violet.**
Everything else — which blue, how many reds, which red — moves.

5 · The deep-red slot

5.1 Swapping only the red part

Six emitters held fixed, one red slot swapped:

red part	I@625	I@660	I@680	far-anchor slope	quiet-window median	source
630–640 nm	★ 0.734	0.359	0.176	⊖ 2.0 %/nm	0.238	measured (Avonec)
★ 660 nm	0.612	0.682	0.196	★ 0.3 %/nm	0.352	★ measured (Avonec)
star 660 nm	0.598	0.686	0.253	0.2 %/nm	0.468	△ modelled
★ star 670 nm	0.598	0.541	★ 0.407	0.2 %/nm	★ 0.545	△ modelled
star 680 nm	0.598	0.401	★ 0.510	0.2 %/nm	0.443	△ modelled
star 690 nm	0.598	0.359	0.390	0.2 %/nm	0.311	△ modelled

⊖ The 630–640 nm part is disqualified, and by the rule the rig already measured. Its rising flank sits inside the 620–630 nm anchor — 2.0 %/nm against the 660's 0.3 — which is the Sansi's documented failure mode, applied deliberately before building rather than discovered after. It buys the best I@625, and pays in exactly the currency the far anchor cannot afford.

5.2 ★ The star is an Avonec part — the second vendor is not needed

The first draft of this study modelled the star as a second-vendor item. **Checking the shop settled it: Avonec lists the 660 nm on a Starplatine themselves.**

bare emitter	3W High Power LED 660nm tiefrot / hyper red — €2.46
★ on a star board	3W High Power LED auf Starplatine 660nm tiefrot / hyper red — €2.75 (€2.29 from 5, €2.15 from 10)
also	the same die on a <i>Grionec-Platine</i> (solderless, plug-together, removable)
stock	ca. 2–4 Werktage on all three
specs listed	3 W, max 800 mA, 2.2–2.8 V, 120° beam, 30–50 lm

⇒ **Two consequences, and both are good:**

★ **1 · The measured spectrum applies.** The star carries the same die as the bare part, so the digitised 660nm.jpg curve — verified to 0.5 nm against the part number — is the *right* input, not a Gaussian stand-in. The whole "unmeasured second-vendor part" risk section of the first draft dissolves.

★ **2 · The star board is what the construction sketch needs.** A 20 mm metal-core PCB is the heat-sink interface for "a white lamp ringed by ~4 small heat-sinked emitters". And **410–420 nm (€3.55) and 430–435 nm (€2.86) are on Starplatine too** — so the entire recommended board is one order, one vendor, one mounting standard.

⚠ **A second vendor becomes necessary only for 670 or 680 nm**, which Avonec does not sell in the 3 W range — and §6 says do not buy those yet anyway.

5.3 How much an unmeasured deep-red bin could move things

Kept as a sensitivity check, since deep-red bins are specified loosely — a part sold as "660 nm" commonly ships anywhere in 650–670 nm:

assumed part	I@625	I@660	I@680	far-anchor slope
650 nm, FWHM 18	0.599	0.545	0.179	0.2 %/nm
650 nm, FWHM 28	0.634	0.618	0.220	⚠ 1.0 %/nm
655 nm, FWHM 22	0.600	0.657	0.210	0.2 %/nm
660 nm, FWHM 22 (nominal)	0.598	0.686	0.253	0.2 %/nm
665 nm, FWHM 28	0.599	0.658	★ 0.377	0.2 %/nm

★ **The far-anchor slope stays at 0.2 %/nm across the whole bin range**, except the widest 650 nm case (1.0 %/nm), the only one whose flank creeps back into 620–630 nm. **Binning risk is low** — the 660 slot is forgiving, and a 665 nm bin is quietly *better* for the quiet window.

5.4 🚫 But no 660 nm part fills the quiet window

	I@680	quiet-window median 660–680
Avonec 660 (<i>measured</i>)	0.196	0.352
star 670	0.407	0.545
star 680	0.510	0.443

A 660 nm emitter is down to ~0.05 of its peak by 680 nm; most of the 0.196 above is the white phosphor's tail, not the LED. **Covering 660–680 properly is a different part number, not a different weight** — and the part is a **670 nm** 3 W star, which Avonec does not stock.

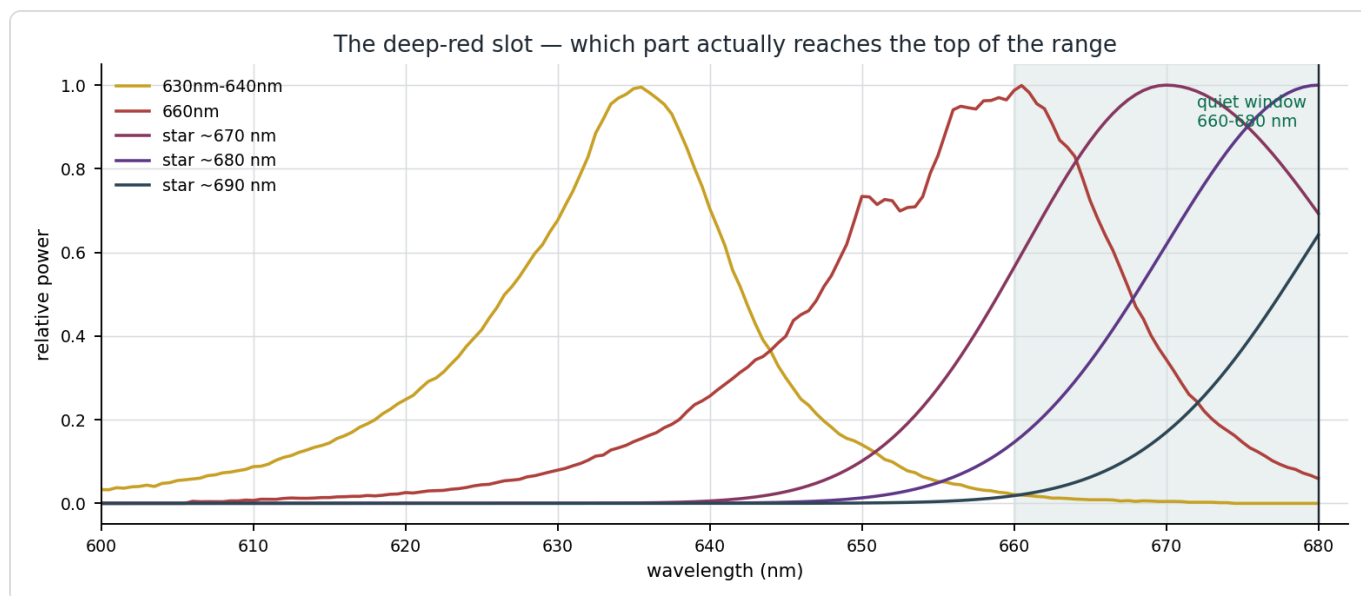


Figure 3 — The deep-red slot. The 630–640 nm part puts its rising flank inside the 620–630 nm far anchor, the failure mode the rig already measured on the Sansi. The Avonec 660 nm part is centred well but is fading by 675 nm. Only a ~670 nm part sits inside the 660–680 nm quiet window with its peak, which is what the "peak where you measure" rule demands.

6 · WITHDRAWN — the claim that the camera ends the range

  This chapter previously argued that response past 640 nm collapses, that the quiet window needs 10×–48× more emitted power than any board can supply, and that the whole thing is gated on the ELP's IR-cut filter. Edwin refuted it on 2026-08-07. The argument is withdrawn, and so are every number that depended on it. What follows is the refutation and what is actually known.

6.1 Why it was wrong

The chapter rested on one figure: `KB_spectroscopy_physics.md` §7.2's *"the red channel falls ~40× between 631 and 657 nm"*.

 That measurement was made through a CFL, and a CFL's red output is Eu^{3+} LINE emission, not a continuum. 631 nm sits on the red flank of the strong 626.6 nm line; 657 nm sits in the gap between 650.7 and 662. A 40× ratio between those two wavelengths is what that lamp's own spectrum does. No instrument roll-off is needed to explain it — and §7.2 itself says the three causes cannot be separated.

 And the rebuttal this document offered was invalid. It compared the 40× against the 631 → 657 fall of phosphor-white LEDs (1.4–1.8×) and concluded the source could not account for it. That is the wrong source class: a white LED has a smooth phosphor hump there, a CFL has lines and gaps. The comparison proved nothing.

★ The measured evidence points the other way. Same camera, §16.25.4: the Sansi clips at 255 DN through 600–640 nm and **still reads 115 DN at 656 nm**. A 40× instrument roll-off between 631 and 657 is not compatible with that — with a phosphor white's own gentle ~1.6× decline it would imply something like 7 000 DN at 631 nm, which is unreachable in 8 bits at a working exposure.

⇒ ★ Working assumption, per Edwin: the optics deliver to ~700 nm and the camera captures it. The spectrometer's range is not known to be limited by the detector anywhere inside 410–680 nm.

6.2 What the camera has actually recorded

The honest replacement for a modelled response curve is the measurement itself — three lamps, as recorded, lamp × instrument together.

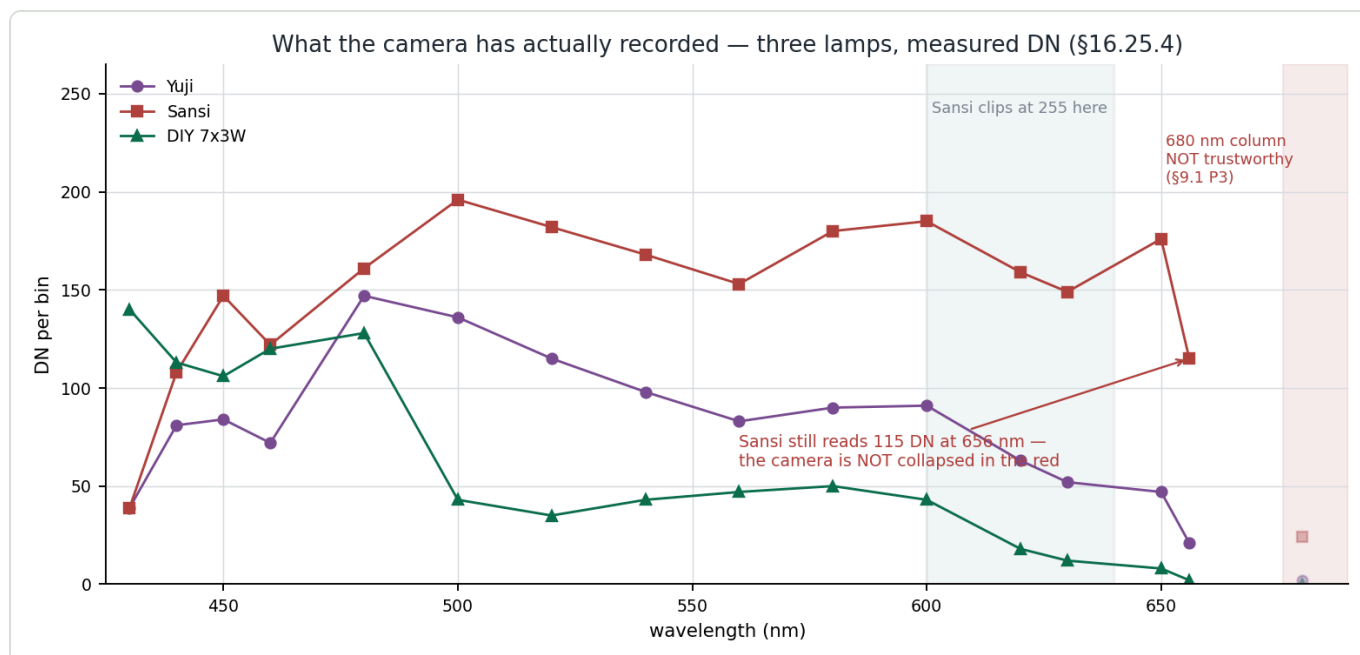


Figure 4 — What the camera has actually recorded: three lamps in DN per bin (§16.25.4). The Sansi holds 149 DN at 630 and still returns 115 DN at 656 nm — the red end is not collapsed. $\triangle$ Two caveats are drawn on the plot: the Sansi clips at 255 through 600–640 nm in the screenshot set, so its mid-range there is a floor rather than a value; and the 680 nm column is dotted because `SPEC_metric_research.md §9.1 P3` records that figure came from “a screenshot ending at ~676 nm with a transferred wavelength scale”.

$\triangle$ **One oddity worth not explaining away:** the Sansi *rises* from 149 DN at 630 nm to 176 at 650. No phosphor continuum does that. Either the two source tables are on different scales, or the Sansi uses a **line-emitting red phosphor** (KSF/PFS, Mn^{4+} , narrow lines near 631/648/660 nm), which would also explain why it owns the deep red. Not decided here.

6.2a A LATER MEASUREMENT CONTRADICTS THIS PANEL — unresolved (2026-08-09; `SPEC_capture_quality.md §16.28.4`)

Two runs captured with the ROI opened to 690 nm — 20260808A (**Sansi V2**) and 20260808B (**Yuji**) — record the red end collapsing on **both** lamps, on the same camera, the same evening:

run	lamp	630	640	650	656	660
20260808A	Sansi V2	54.1	21.0	1.2	0.15	0.07
20260808B	Yuji	39.7	11.5	0.7	0.13	0.01

Against this chapter’s “*still returns 115 DN at 656 nm*” that is a two-orders-of-magnitude disagreement in **shape**, and both records cannot be right. $\triangle$ The **Yuji** rows are consistent between the two datasets (52 against 39.7 at 630 nm — the same ballpark at a different exposure). **The Sansi is where they diverge.**

$\triangle$ **This does NOT restore the withdrawn §6 argument.** Three reconciliations remain open, and the 2026-08-09 session cannot choose between them:

- V1 $\neq$ V2.** §16.25.4’s Sansi is the V1; this is the V2. §6.2’s own KSF/PFS hypothesis would explain a V1 that owns the deep red and a V2 that does not. $\triangle$ Though the new runs show **no line structure** at 648 or 660 on either lamp — a smooth monotone decay, which is what a KSF lamp would *not* produce.
- The red end of Figure 4 is mis-scaled** — this chapter already flags that its 680 nm column came from “a screenshot ending at ~676 nm with a transferred wavelength scale”. That is exactly the failure mode that relabels a 630 nm value as “656 nm”.

3. **Figure 4's mid-range is clipped** at 255 DN through 600–640 nm, so its red-end *shape* is not comparable with an unclipped run.

★★ **What decides it, with hardware already on the bench:** EUROPIUM_RED_FAR_680/690/700 (687.7, 693.7, 707.0 nm) are the Eu³⁺ siblings of the 611.6 nm line the calibration already uses, and they are strong in the calibration lamp — never markable before because the ROI ended blue of them. **Lines visible ⇒ the camera passes 690 nm, §6's withdrawal stands, and the collapse belongs to the lamps. Lines absent ⇒ the IR-cut is the gate and §6 must be re-opened.** Queued, not concluded.

★★ 6.3 The argument that replaces it — the 619/624 nm red feature

Withdrawing §6 does **not** leave the lamp choice unsupported at the red end. There is a real, measured effect there, and it is a better argument than the one it replaces because it rests on rig data rather than on a model.

SPEC_capture_quality.md §16.26.11 records it:

what	a sharp absorbance feature at 619/624 nm that moves between runs
not noise	200–600× the point-to-point standard deviation
★ not present in nulls	invisible in every same-liquid null (615–622 nm reads $+0.0011 \pm 0.0009$) — so it needs <i>something to change between R and S</i>
★ present on BOTH lamps	archive Yuji 0.008–0.013 against Sansi 0.046–0.078 — 4–6× larger in <i>absolute</i> terms, so not merely the weaker fills

⇒ ★ **The Sansi's 619 nm edge (–11 %/nm; §16.25.4 measures 25 %/nm at 622.0) AMPLIFIES the feature 4–6×. It does not cause it.** The cause is still unidentified and requires something to change between the two captures; the **refill null** (§16.26.11) is the pre-registered run that would find it, and it has not been run.

⚠ **Correcting a premise while we are here: the archive lamp was the Yuji, never the Sansi.** §16.26.9 classifies the archive runs by blue/red ratio at ≈ 2.97 — the Yuji. The Sansi appears only as a candidate in the 2026-08-06 null and probe runs. **No Sansi structure can have affected the measurements the metric and its thresholds were built on.**

⚠ **And a second correction, from 2026-08-09: the 25 %/nm-at-622 disqualifier is V1-SPECIFIC.** Measured on the **Sansi V2** (20260808A), the far anchor reads **1.47 %/nm median, 3.05 %/nm max** — Yuji-class — and the V2's sharp feature has moved to **614.3 nm**, *outside* the anchor, structurally the same situation as the Yuji's 610.9 nm line. ⇒ By this document's own rule — **it is LOCATION, not steepness** — the V2 passes the test the V1 failed, and the recorded "do not adopt the Sansi" should be read as applying to the V1.

⚠ **The V2 was nonetheless NOT adopted** (SPEC_capture_quality.md §16.28.7). Measured against the Yuji on the same rig it carries the white-LED valley between blue pump and phosphor — $I(470)/I(600) = 0.35$ against the Yuji's **1.46** — and reproduces the V1's Soret steepness (7.48 %/nm across 440–460 against the Yuji's 2.27). ★ The decision rests mainly on **heat**: the V2 draws ~30 W against the Yuji's ~20 W, and §16.11.16 measured a warmed fill reading as a *browner oil*, misclassifying 3 of 3 runs, with §16.11.17's decay-rate run queued next.

★★ **And this is the practical case for R2's flatness.** The feature lands squarely on the **far anchor 620–630**, the band carrying §16.24.2's **17× leverage**. R2 reads **0.1 %/nm at 625 and 0.2 %/nm across 620–630**, against the Sansi's 11–25 %/nm. ⇒ **the recommended lamp removes the multiplier.** It cannot remove the

disturbance — nothing about a lamp can, since it happens *between* the two captures — but 4–6× off an error in the highest-leverage band is the single most concrete benefit in this document.

⚠ **Keep this separate from a different artefact in the same neighbourhood:** the archive is **convex at 626 nm on every fill**, the Qy maximum has never been observed, and a null-run control showed the **instrument's own curvature** would fake one (SPEC_metric_research.md §9.1 item 5). Same region, not the same effect, and it is not attributed to lamp line structure.

6.4 What survives, and the one test still worth running

⛔ withdrawn	"past 640 nm the camera is the limit"; the 10×–48× parity figures; the delivered-score ranking; the advice to hold the deep-red purchase
✓ untouched	§4 and §5 — the emitted-spectrum ranking and the R2 build. Neither ever used the response model
★ still worth doing	the one-evening test below, now as a <i>characterisation</i> rather than a gate

★ **The test, and it needs no datasheet, no absolute calibration and no known flux:**

Shine the Avonec 660 nm LED alone into the slit and capture it. What the instrument records is $SPD(\lambda) \times response(\lambda)$. The $SPD(\lambda)$ is known — the digitised 660nm.jpg curve of §3.1, verified against the part number to 0.5 nm. **Divide the captured spectrum by it, and the quotient is the instrument's response across ~640–690 nm**, up to one unknown constant, which cancels because only the shape is in question.

★ **Chain it with the 630–640 nm part.** That curve spans ~605–665 nm, so the two overlap across ~645–665 nm; matching them there stitches one response curve from **605 to 690 nm** with no absolute measurement anywhere — and the same capture independently validates the digitised SPDs.

⇒ It is no longer a purchase gate. It is the measurement that would have prevented this chapter's error, and both parts are on the R2 order anyway.

7 · Recommendation

7.1 ★★ The board to build — R2, and every part is Avonec

3 × Avonec 4000–4500 K + 2 × Avonec 410–420 nm + 1 × Avonec 430–435 nm + 1 × Avonec 660 nm — all four part numbers available *auf Starplatine*, one order, one vendor, ~€20 in emitters.

Its spectrum is the **solid green curve in Figure 2**.

slot	part	why
backbone	3 × 4000–4500 K	★ not 6500 K — beats every other white on this range (50.12 vs 55.71) and has the best red tail of the five
violet	2 × 410–420 nm	the Soret bracket, and the biggest single gain in the study
blue	1 × 430–435 nm	sits nearly on the Soret peak — I@432 0.810 against 0.670 for the 440–450 nm part

slot	part	why
deep red	1 × 660 nm	★ the measured Avonec part; far-anchor slope 0.3 %/nm, and it triples I@660 (0.353 → 0.682)
green	— none	the 4000 K phosphor already gives 0.89 at 540 nm. This reverses the 430–670 study
cyan 480	II only at 8 slots	fills the 476 nm hole 3.6× but costs half the Soret bracket at seven (§4.2)

7.2 If only one part is bought

★★ 1 × Avonec 3 W 410–420 nm, on the existing lamp, behind the existing diffuser. On a bare 4000–4500 K board it lifts 410 nm **16×** and the sub-Soret level **3.3×**, taking the score from 50.12 to 26.46 — the largest single gain here, it survives all five weightings including the control that deletes the criterion it was scored on, and it needs no new board.

△ Its benefit is **invisible until the capture clamp moves below 440 nm** — but that clamp is a software choice, not a hardware one.

△ And "one part" understates the work. §7.1 requires every emitter behind the **same diffuser with working distance**, or the Yuji's 1.8 % along-slit uniformity is traded away — a bare point emitter beside a diffuse *panel* lamp **selects** emitters rather than mixing them. Getting those two behind one common diffuser is arguably **harder** than the co-planar seven-emitter board, where the geometry solves itself. ⇒ read it as *one part plus a mixing problem the full build does not have*. §7.4 offers a way round it.

7.3 The 670 nm reach — re-priced

△ This section previously said "do not buy the 670 nm reach yet", on §6's now-withdrawn argument. With that argument gone, so is the reason to wait.

★ On the evidence available, a 670 nm 3 W star is straightforwardly worth having. The quiet window 660–680 nm is the only pigment-free region in the whole range (§2.2), it is the baseline anchor the metric has never had, and §8's forecast puts the oil's absorbance there at **A ≈ 0.008** — the light that arrives is the lamp, undiminished.

△ The two costs that remain are real but small:

△ a second vendor	Avonec sells no 670 nm in 3 W. It is the only part on the board that needs one
△ it is modelled	no measured SPD; §5.3 shows the slot is forgiving, but the curve is a skew-normal, not a measurement

⇒ Sequence:

	step	cost
1	★ Order R2 — all Avonec, all on Starplatine	~€20 in emitters
2	★ Add a 670 nm star from a second vendor if the quiet window is wanted now, or defer it as the one part that needs a second source	~€3–5

	step	cost
3	★ Run the 660 nm response test (§6.4) with a part from that same order — now a characterisation, not a gate	one evening
4	Move the capture clamp below 440 nm and above 630 nm and confirm both new ends	free — software

△ **And whatever is built, build it as one rig change.** Every one of these moves invalidates the thresholds, and the recalibration is the expensive part — bundling them means paying for it once.

★ 7.4 De-risk both halves before building anything

Neither half of the case has to be taken on trust. Both can be tested before a board exists.

(a) ★ The red half is free today — no hardware at all. Move the capture clamp past 630 nm on the **existing Sansi**, which already reads 149 DN at 630 and 115 at 656 (§6.2), and look: does 660–680 nm behave as a flat, pigment-free anchor, and does `far/near < 1` appear? That is a calibration and a software clamp, and it tests the single most valuable claim in this study — §2.2's quiet window — before anything is ordered.

(b) The blue half needs one emitter, but not bolted beside the Yuji. §7.2's caveat applies: a point source next to a diffuse panel is a mixing problem. ★ **Cleaner: capture with the violet ALONE.**

$T = S/R$ only needs the lamp to have light **where you are measuring**, and both captures use the same lamp. So a violet-only source across ~405–450 nm yields a valid transmission curve over the whole Soret, and answers the one question the blue end exists for: **does the band's position move between a green and a brown oil?**

✓ No integration with the Yuji, no common diffuser, no uniformity regression — a lamp swap for one probe capture. 🚫 **It cannot produce M** : the baseline chord spans 510–540 and 620–630 and a violet-only lamp has neither. It is a **shape probe, not a verdict**. △ Still needs the calibration extended below 440 nm, and its own exposure handling.

⇒ **Together these price both halves of the thesis for one calibration session and one emitter.**

8 · ★ What it would boil down to — a green-vs-brown forecast

(Edwin 2026-08-07: "simulate an expected absorption curve based on typical green and brown oils ... just an anticipated estimate telling us what it all would boil down to".)

△△ A FORECAST, NOT A MEASUREMENT. No oil was measured for this. It is a forward model of $A_{oil}(\lambda)$ from the pigment physics of KB_spectroscopy_physics.md §4.1/§4.1a, with two free parameters tuned to values the rig actually reports. It can say **where the light goes and which regions are usable**. It cannot confirm any band ratio, because the ratios were put in by hand.

8.1 What it is anchored to

anchor	value	source
band centres	432 / ~574 / ~625 nm	protochlorophyll, KB §4.1a
the shipped windows	Soret 440–460, near 510–540, Q 560–580, far 620–630	DevSpectralPlugin.py
★ the class values	M baseline green 15.559 , brown 10.160 , threshold 10.35	SPEC_roast_ampel.md §2b, 37 runs
★ the absolute level	B_Soret (440–460) = 1.0272	SPEC_metric_research.md §7.2
the green → brown direction	Qy −17 %, 572 nm +14 %, per unit Soret	§16.11.16, one demetallated fill

△ Both anchors are needed, and the second is easy to miss: M is a ratio of two baselined band means, so it is invariant under an overall scaling of A. Fitting M alone leaves the absorbance level free — exactly the quantity every photometric question here depends on. B_Soret pins it.

8.2 △ The one falsifiable check — and it fails

Both free parameters went on M and on the 440–460 band level, so the **448–460** window was not fitted. Its ratio to 440–460 is therefore a genuine prediction of the Soret slope's shape:

model	0.942
rig	0.674 (B_Soret 0.6924 / 1.0272)
	△ 40 % apart

Trimming 440–447 off costs the rig **33 %** of its Soret band and costs the model **6 %**. So the real curve rises far more steeply below 448 nm than a Soret centred at 432 with a 42 nm width can. Two readings:

(a) the model's Soret is too broad or mis-placed — in which case the blue-end story needs re-checking, since it rests on that slope;

(b) ★ the rig's **440–447 bins are not real absorbance**. §7.13 documents exactly this: they are starved bins, and $A = -\log_{10}(S/R)$ blows up as S approaches the floor, which inflates the 440–460 mean. SPEC_metric_research.md §7.2 calls the trim "drops the non-measurements" — the same claim from the metric's side.

⇒ (b) is independently documented and predicts the sign and rough size of what is seen, so the failure is consistent with a known instrument artefact rather than evidence against the pigment model. **➡ But it is not proof, and it leaves the model's Soret slope unvalidated.** Every blue-end number below is indicative.

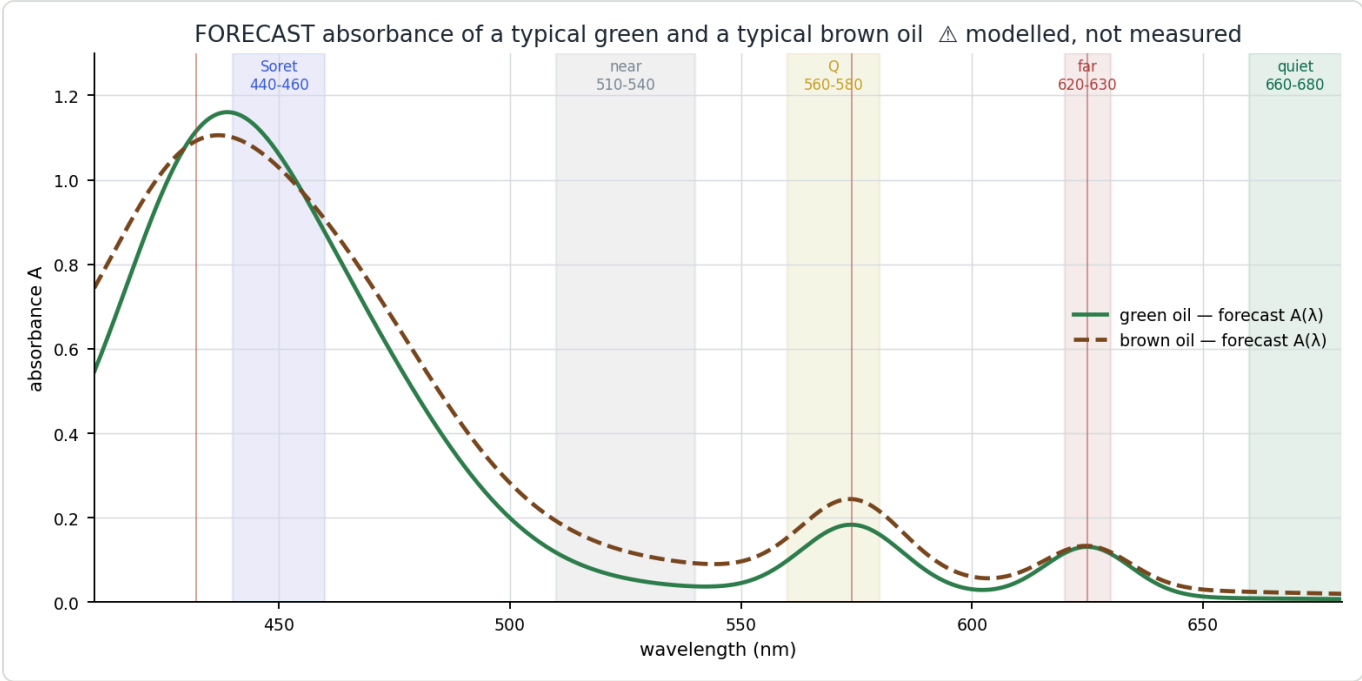


Figure 5 — Forecast absorbance of a typical green and a typical brown oil. Δ Modelled, not measured. The brown curve is blue-shifted (demetallation), raised in the Q band and lowered at Qy — the directions the rig measured on an aged fill. Note the composite blue maximum sits near **440 nm, not at the Soret's 432 nm centre**: the carotenoid absorption at ~455 nm rides on top of the Soret, so the peak you can see is not the peak the chemistry is at.

8.3 What the camera would see

Transmitted signal $S = R \cdot 10^{(-A)}$, relative to the lamp's own peak. **Below 1 % is the starved regime that made 440–447 dead bins.**

	R2 lamp	S green	S brown	incumbent	S green (incumbent)
410 nm	0.442	★ 0.126	0.080	➡ 0.036	Δ 0.010
432 nm — Soret peak	0.810	0.062	0.066	0.815	0.063
450 nm	0.561	0.049	0.053	1.000	0.087
520 nm	0.572	0.486	0.416	0.789	0.670
574 nm — Q	0.893	0.585	0.509	0.779	0.511
625 nm — Qy	0.613	0.453	0.451	0.393	0.291
660 nm	0.682	★ 0.667	0.644	0.485	0.475
680 nm	0.196	0.193	0.187	Δ 0.099	0.098

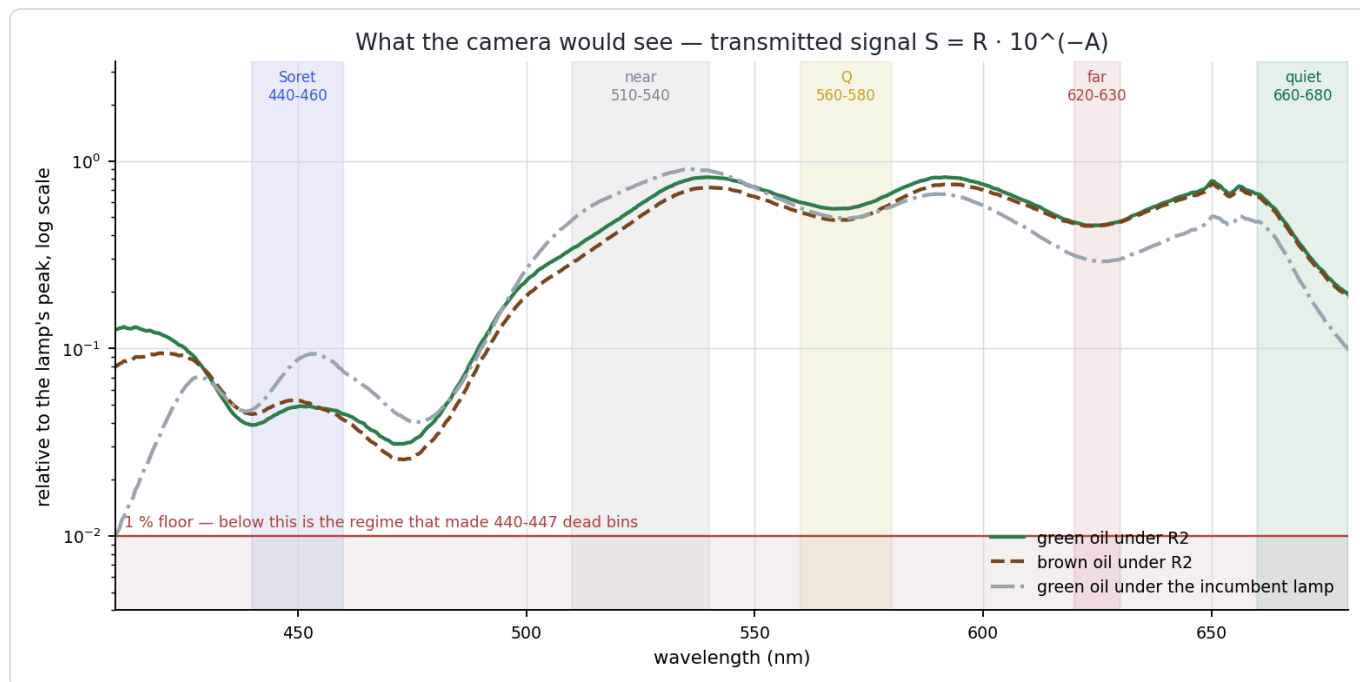


Figure 6 — What the camera would see. Green and brown under the recommended R2 lamp, against green under the incumbent. The five measurement windows are shaded. Nothing falls below the 1 % floor anywhere in 410–680 nm under R2 — which is the headline of this chapter.

8.4 ★ What it boils down to

1 • The quiet window is the clean win. $A(660-680) = 0.008$ green / 0.022 brown — effectively transparent, so what arrives there *is* the lamp. R2 delivers 0.682 at 660 nm against the incumbent's 0.485 , and **0.196 vs 0.099 at 680 nm ($2.0\times$)**. $\Rightarrow$ a baseline anchor with no pigment in it, which $620-630$ is not and never was.

2 • ★ The blue gain is BELOW the peak, not at it — and that is the point. At 432 nm the two lamps are level (0.810 vs 0.815): the incumbent already carries two $430-435$ emitters. The difference is underneath:

	R2 lamp	incumbent	gain	S green (R2)	S green (incumbent)
410 nm	0.442	0.036	★ $12.4\times$	0.126	△ 0.010 — <i>at the floor</i>
415 nm	0.628	0.090	$7.0\times$	0.129	0.019
420 nm	0.823	0.241	$3.4\times$	0.119	0.035
425 nm	0.963	0.571	$1.7\times$	0.102	0.060
430 nm	0.899	0.835	$1.1\times$	0.074	0.069

$\Rightarrow$ ★ **The incumbent cannot see the Soret's short-wavelength side, so it cannot see the demetallation blue shift. R2 can.** That is the entire case for $410-420$ nm, and it is a *shape* argument, not a photon-count one.

3 • △ The blue is expensive in signal either way. $A(432) = 1.11$, so the oil transmits **7.7%** there at the working recipe. Under R2 that is 0.062 of the lamp peak — above the floor, but not comfortably. Halving the concentration would gain **$3.6\times$** . $\Rightarrow$ **the lamp and the capillary protocol are one change, not two.**

4 • ★ The three peaks do sit on a smooth lamp — which is what you expected. $|d\ln I/d\lambda|$ of R2:

432 nm Soret	6.0% /nm	on the $430-435$ emitter's own peak
---------------------	-------------	-------------------------------------

574 nm Q	2.0 %/nm	on the 4000 K phosphor's smooth hump
625 nm Qy	★ 0.1 %/nm	flat — the 660 part's rise starts later

For scale: an emitter **edge** inside a band costs 25 %/nm (the Sansi's measured failure, §16.25.4); a source **centred** on a band costs 0–5 %/nm. All three land in the good regime, and the Qy anchor — the one §16.24.2 says carries 17× the leverage — is the flattest of the three.

5 • △ **A prediction that can be falsified.** The S1 window trim 440–460 → 448–460 should move **M** from 15.56/10.16 to **14.66/9.72**, class gap 53 % → 51 %. △ Believe the direction, not the number — the brown oil's chemistry is modelled from **one** measured fill (§16.11.16), which is the thinnest input here.

★★ 8.5 What the complete bands open (Edwin 2026-08-07)

SPEC_metric_research.md §9.1 closed **seven** analysis routes in a single session, *"every one of them on the same wall: both principal bands are flanks and the range is clipped at both ends."* R2 plus the clamp move is what removes that wall:

band	today	with R2 + the clamp
Soret 432	right flank only (448–460); no light below 430	★ a complete band — 0.44 at 410, 0.81 at 432, both flanks visible
Q 574	already contained	unchanged, better lit (0.89)
Qy 625	at the clamp edge, used as a <i>baseline anchor</i>	★ a complete band , with a pigment-free region beyond it

△ **One asterisk on "complete"**: the carotenoid at ~455 nm rides on top of the Soret, so even with full coverage the observed blue maximum is a **composite** — §8.2's model puts it near 440, not 432. Separating them needs decomposition, which is the first technique the full band enables. *The band you can only get by covering it fully is also the band you then need to deconvolve.*

What becomes possible:

	technique	why it needs the full band
1	★★ peak-POSITION metrics	a position is dilution-invariant by construction — Beer–Lambert scales amplitude, it does not move a maximum. That is the target property the metric research states and no ratio has . The demetallation blue shift is a position, and it is unreachable from one flank
2	band decomposition (C7)	fitting Gaussian/Voigt shapes needs the whole band; returns position, width and area separately — three numbers where today there is one. §9.1 called it <i>"justified only if R2/R3 leave the truncated peaks as the limiting factor"</i> . They did
3	derivative spectroscopy (C20)	2nd derivatives resolve overlapping bands and are baseline-free. §9.1 ranked it last because <i>"its weakest term is at the range edge"</i> — move the edge and that objection dies
4	a real baseline	anchoring on the pigment-free 660–680 window instead of a chord through 620–630, which <i>contains the Qy band</i> . Would make r_Q , the pedestal correction and assumption A1 unnecessary rather than unproven

	technique	why it needs the full band
5	moments (C19), three-band (C18), and the scatter-correction family (§9.1 P3, via $\text{far/near} < 1$)	all need the region beyond the bands

🚫 **The gate that comes with it, and it is unclosed.** Position metrics inherit `SPEC_metric_research.md`
R1b: *is λ stable to $\ll 0.9$ nm across sessions?* Never verified. A blue shift of a few nm measured against a calibration that may wander by 0.9 nm is not a measurement. ★ Cheap to close — the CFL gives **Hg 404.66 / 435.83 / 546.07 / 576.96 / 579.07 plus Eu³⁺ 611.6 / 626.6 / 650.7** in a single capture, an eight-point check across the whole working range. Worth doing in the same evening as §6.4's response test.

⚠️ **And the honest caveat on the whole chapter: the lamp is not today's bottleneck.** The null series put the instrument at **0.42 %** and reseating at **~1.4 %** against a 3–5 % archive CV; §16.26.10 concluded the residual **~3.8 % is the preparation**. A better lamp does not touch that term. ⇒ This is **access to a class of measurement that is currently impossible**, not a reduction of today's noise — better *in kind*, and untested.

9 · What would change these conclusions

#	if this turned out otherwise	what changes
1	🚫 the instrument response past 640 nm — the claim that it collapses is WITHDRAWN (§6.1); the 40× was the CFL's own line structure	already applied: §7.3 re-priced, §6's parity figures deleted. §6.4's test remains worth running, now as characterisation
1b	⚠️ the model's Soret slope — §8.2's shape check fails by 40 %, explained by a documented artefact but not proved	§8's blue-end numbers, which are indicative only. The <i>ranking</i> in §4 does not use the oil model at all
2	★ per-part radiant flux — assumed equal across 3 W parts	the <i>counts</i> , not the <i>parts</i> . The 4000 K backbone and the 410–420 violet survive any plausible reweighting
3	the 660 nm bin — assumed nominal	little; §5.3 shows the slot is forgiving across 650–670 nm, with only a wide 650 nm bin creeping into the far anchor
4	diffuser transmission (40–80 %) and along-slit uniformity	not modelled at all. Every emitter must sit behind the same diffuser with working distance, or the 1.8 % uniformity bar is traded away
5	the digitised curves	verified peak-by-peak against the part numbers (§3.1) to 1–6 nm
6	⚠️ the catalogue	this study was wrong once already about what Avonec sells (§1.1). The 480–485 nm part was missed; the harvested SPD set should be re-checked against the live shop before the order goes out

⚠️ **The one thing this study cannot do is see the benefit it recommends.** With the capture clamp at 440–630 nm, both new ends are outside the measurement. The lamp change and the clamp change are one project, not two.

10 · Reproducing this

```
cd spectracsPy
./venv/bin/python diagnostics/led_lamp_410_680.py --verify --figures
PYTHONPATH=diagnostics ./venv/bin/python diagnostics/oil_forecast_410_680.py --figures
```

`--verify` re-derives every digitised peak and checks it against the part number; `--figures` writes Figures 1–4 to `spectracs-references/tmp/lamp410680/`, and the forecast script writes Figures 5–6 to the same place. The full ranking goes to `ranking.json`.

Related: `SPEC_capture_quality.md` §16.25.4 (the lamp brief), §16.25.4a (the 430–670 nm study this widens), §16.25.4b (this study's summary) · `KB_spectroscopy_physics.md` §4.1 (the pigment correction), §4.1a (the three band centres), §7.2 (the delivered range) · `KB_led_and_oil_spectra.md` (the Avonec catalogue and the harvest method) · `SPEC_metric_research.md` §9.1 item 5 (extend the red, light the blue) · `diagnostics/led_combination_search.py` (the 430–670 nm study).